

Reviewer Pack — Minimal Rank of Configuration Spaces Bounds Monomial Ideal C...

Entry e1f654064802 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Configuration Spaces Bounds Monomial Ideal Complexity

Entry ID: e1f654064802 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-28 21:02:20 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Topology (Configuration Spaces) **Field B** (complexity object): Complexity Theory (Monomial Ideal Complexity)

Statement:

[‘For a given n -vertex graph G , let $I(G)$ be the monomial ideal associated with its edge set. The minimal rank of the configuration space of G , denoted as $\text{Rank_conf}(G)$, is upper-bounded by the Kronecker dimension of $I(G)$, i.e., $\text{Rank_conf}(G) \leq \kappa(I(G))$.’, ‘For all n -vertex graphs G , $\kappa(I(G)) = O(\log(n))$.’, ‘There exists a constant $c > 0$ such that for every graph G with at least n vertices, $\kappa(I(G)) \geq c \cdot \text{Rank_conf}(G)$.’]

Rationale (proposer’s reasoning):

[‘Configuration spaces in algebraic topology have previously been used to study geometric properties of graphs. Monomial ideals, on the other hand, are closely related to computational complexity issues like circuit complexity and communication complexity.’, ‘This conjecture suggests a potential connection between the geometric structure of graphs (configuration space) and their combinatorial representation (monomial ideal).’, ‘If true, it could provide a new insight into the relationship between geometric and algorithmic properties of graphs.’]

Taxonomy category: CONFIG_SPACE_MONOMIAL Ideal (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 5e4170cf919d05ae

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For all graphs G with at least n vertices, if $\kappa(I(G)) \leq c \cdot \text{Rank_conf}(G)$, where $\kappa(I(G))$ is the Kronecker dimension of $I(G)$ and $\text{Rank_conf}(G)$ is the minimal rank of the configuration space of G , then support for the conjecture is provided. Falsification occurs when $\kappa(I(G)) > c \cdot \text{Rank_conf}(G)$ for any graph G .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	1.00	SAFE	SAFE
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "configuration space" AND "monomial ideal complexity" - "minimal rank of configuration spaces" AND "Kronecker dimension" - "edge set graph" AND "upper bound Kronecker dimension"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=1.3s

5.1 Generated Python source

```
import random
from fractions import Fraction
from itertools import combinations

def random_graph(n, m):
    edges = set()
    while len(edges) < m:
        u, v = random.sample(range(n), 2)
        if u != v and (u, v) not in edges and (v, u) not in edges:
            edges.add((u, v))
    return edges

def kronecker_dimension(edges):
    n = max(max(u, v) for u, v in edges) + 1
    A = [[Fraction(0, 1) if i != j else Fraction(1, 1) for j in range(n)] for i in range(n)]
    for u, v in edges:
        A[u][v] = Fraction(0, 1)
        A[v][u] = Fraction(0, 1)

    rank = 0
    for row in A:
        if any(x != Fraction(0, 1) for x in row):
            pivot_col = next(j for j, x in enumerate(row) if x != Fraction(0, 1))
            for i in range(n):
                if i != rank and A[i][pivot_col] != Fraction(0, 1):
                    factor = -A[i][pivot_col] / A[rank][pivot_col]
                    for j in range(n):
                        A[i][j] += factor * A[rank][j]
            rank += 1
    return rank
```

```

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n_values = [5, 10, 15, 20, 30, 40]
    results = []

    for n in n_values:
        instances_tested = 0
        total_kronecker_dim = 0
        total_rank_conf = 0

        for _ in range(6): # Ensure at least 30 instances per seed
            edges = random_graph(n, n)
            kronecker_dim = kronecker_dimension(edges)
            rank_conf = len(list(combinations(range(n), 2))) - len(edges) # Simplified Rank_conf

            total_kronecker_dim += kronecker_dim
            total_rank_conf += rank_conf
            instances_tested += 1

        mean_kronecker_dim = total_kronecker_dim / instances_tested
        mean_rank_conf = total_rank_conf / instances_tested

        results.append((mean_kronecker_dim, mean_rank_conf))

    metric_name = 'Kronecker Dimension vs Rank Conf'
    metric_value = sum(kronecker_dim * rank_conf for kronecker_dim, rank_conf in results) / len(results)
    conjecture_holds = all(kronecker_dim <= 10 * rank_conf for kronecker_dim, rank_conf in results)
    counterexample = 'if conjecture_holds else 'mapping_undefined'

    return {
        "metric_name": metric_name,
        "metric_value": metric_value,
        "instances_tested": len(results) * instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_metric_value = sum(result["metric_value"] for result in results) / len(results)
    std_metric_value = (sum((result["metric_value"] - mean_metric_value) ** 2 for result in results) / len(results)) ** 0.5
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")

```

```

elif any(not result["conjecture_holds"] for result in results):
    first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_failing_seed}")
else:
    print("RESULT: INCONCLUSIVE reason=unsupported_operation")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

'Kronecker Dimension vs Rank Conf', 'metric_value': 7767.5, 'instances_tested': 36, 'conjecture_holds': True
TRIAL: {'metric_name': 'Kronecker Dimension vs Rank Conf', 'metric_value': 7766.944444444445, 'instances_tested': 36, 'conjecture_holds': True}
TRIAL: {'metric_name': 'Kronecker Dimension vs Rank Conf', 'metric_value': 7790.694444444444, 'instances_tested': 36, 'conjecture_holds': True}
TRIAL: {'metric_name': 'Kronecker Dimension vs Rank Conf', 'metric_value': 7773.472222222223, 'instances_tested': 36, 'conjecture_holds': True}
TRIAL: {'metric_name': 'Kronecker Dimension vs Rank Conf', 'metric_value': 7737.5, 'instances_tested': 36, 'conjecture_holds': True}
TRIAL: {'metric_name': 'Kronecker Dimension vs Rank Conf', 'metric_value': 7786.25, 'instances_tested': 36, 'conjecture_holds': True}
TRIAL: {'metric_name': 'Kronecker Dimension vs Rank Conf', 'metric_value': 7723.75, 'instances_tested': 36, 'conjecture_holds': True}
TRIAL: {'metric_name': 'Kronecker Dimension vs Rank Conf', 'metric_value': 7810.0, 'instances_tested': 36, 'conjecture_holds': True}
TRIAL: {'metric_name': 'Kronecker Dimension vs Rank Conf', 'metric_value': 7807.777777777778, 'instances_tested': 36, 'conjecture_holds': True}
TRIAL: {'metric_name': 'Kronecker Dimension vs Rank Conf', 'metric_value': 7791.944444444445, 'instances_tested': 36, 'conjecture_holds': True}
TRIAL: {'metric_name': 'Kronecker Dimension vs Rank Conf', 'metric_value': 7788.333333333333, 'instances_tested': 36, 'conjecture_holds': True}
TRIAL: {'metric_name': 'Kronecker Dimension vs Rank Conf', 'metric_value': 7804.305555555556, 'instances_tested': 36, 'conjecture_holds': True}
RESULT: SUPPORTED mean=7778.657407407407 std=21.90544863038005 support_fraction=1.0

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The empirical test only includes 36 instances, which is too small to draw a definitive conclusion about the conjecture. The metric may not scale trivially with n , and there could be larger graphs where the conjecture does not hold.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge | original judge: The test results indicate that for all tested instances, $\kappa(I(G)) \leq c \cdot \text{Rank_conf}(G)$ holds true, which supports the conjecture. | next: Further investigation is needed to confirm the conjecture's validity for larger graphs. Test more diverse and larger graphs to ensure the metric scales as expected.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11922
2	preregistration	ollama_remote	glm4:latest	0	0	5630
3	novelty	ollama_remote	glm4:latest	0	0	4592
4	novelty	ollama_remote	glm4:latest	0	0	6745
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17271

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13291
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10246
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12115
9	critic	ollama_remote	glm4:latest	0	0	9245
10	judge	ollama_remote	glm4:latest	0	0	5600

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 96656 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in *research/audit/elf654064802.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/elf654064802.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/elf654064802.tar.gz* (if generated)